

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0490

Amelia St 100 kV
July 24, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0490. The Interconnection Customer proposes a 70.7 MW battery energy storage facility interconnecting at the Amelia St 100 kV substation in the Indiana area. The request was received on June 23, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 278 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$811,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Lansing - Ucon 100 KV Line 1
Contingency: Lansing - Ucon 100 KV Cir. 2
Loading: 106.3% (Rating: 146 MVA)
DFAX: 34.0%
Pre-project loading: 89.8%
- Facility: Lansing Substation - Ucon 100 KV
Contingency: Lansing - Ucon 100 KV Ckt 1
Loading: 106.3% (Rating: 146 MVA)
DFAX: 34.0%
Pre-project loading: 89.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lansing - Ucon 100 KV Line 1	Reconductor line	\$435,000	70% (\$303,000)
Lansing - Ucon 100 KV	Reconductor line	\$731,000	70% (\$508,000)
	Total allocated to project		\$811,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.